OS-Helix: Unifying GUI Interaction and Code Execution in Native Computer-Using Agents

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Paper version: arXiv 2610.22234v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Real desktop automation is neither purely visual nor purely programmatic.*

A bolt-on code tool is not enough: **34.17** → **32.68** on OSWorld (Qwen3-VL-32B, Table 14).

Motivation: three gaps

1. **GUI-only.** Small targets, clumsy structured edits.
2. **Code-only.** Blind to layout and UI state.
3. **Bolt-on code.** No routing; GUI score can drop.

Evaluation setup

OSWorld-Verified · 361 tasks | WindowsAgentArena | Qwen3.5-27B / 35B-A3B | Sonnet 4.6 teacher | 50-step budget | RL: 1,000 steps, 27B

- **All online** on live VMs; offline grounding only in controls.
- **Held out?** No app split: synthesis covers OSWorld apps; domain removal is the nearest control.

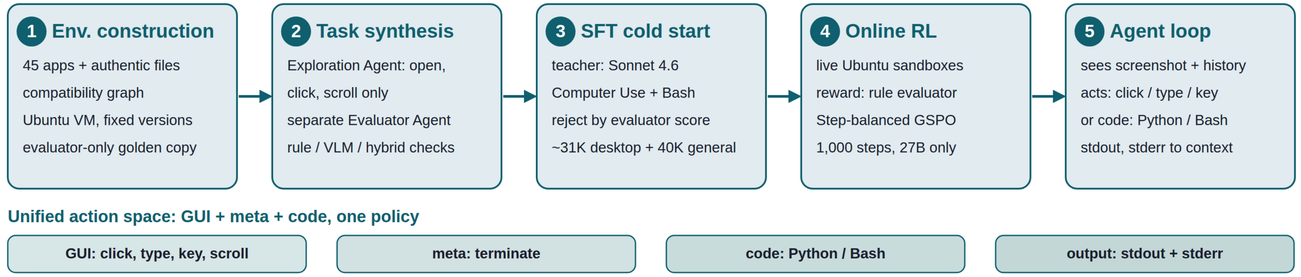
Our read (CUA Radar)

75	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; domain removal; Pass@3 bootstrap CI; matched baselines; ablations.
- **Flags:** synthesis overlaps eval apps (AP5). **Not reported:** evaluator validation; code.
- Pattern C11. Authors' numbers, not rerun.

Method: one policy, one hybrid action space, verifiable synthesized tasks

- **Sees / acts:** screenshot (up to 8 past) + text history in; GUI action or Python / Bash out.
- **New:** app–artifact states, decoupled instruction and evaluator agents, SFT then RL.
- **Signal:** rule-based evaluator score is the sole RL reward; ratio averaged per action step.

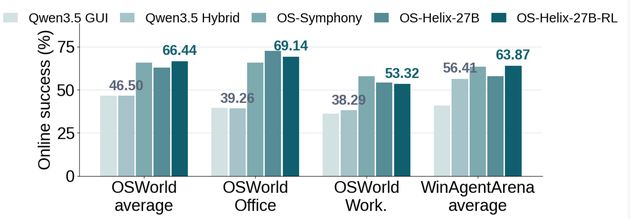


Our redraw (not the paper's figure): the four training stages, the agent loop and the unified action space.

$$w_i^{\text{step}}(\theta) = \exp\left(\frac{\sum_{k=1}^{K_i} \alpha_{\tau_{i,k}}\, \bar{\delta}_{i,k}}{\sum_{k=1}^{K_i} \alpha_{\tau_{i,k}}}\right)$$

Step-balanced GSPO: mean log-ratio per action step, GUI and code weighted equally

Key results: online success rate (%)

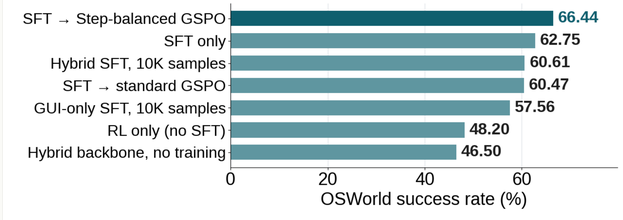


Tables 1, 2 redrawn: Qwen3.5 = 27B base (repro.); OS-Symphony w/ GPT-5; Work. = multi-app; 50 steps.

OSWorld: 62.75 SFT, 66.44 RL vs 46.50 Hybrid baseline

SO WHAT → *Larger newer models (Kimi K2.6 73.07, Qwen3.7-Plus 73.33) score higher.*

Ablation: OSWorld success rate (%), Qwen3.5-27B



Tables 4 and 5, redrawn: one change at a time; 10K rows use a 10K-sample SFT budget.

SO WHAT → *Standard GSPO falls to 60.47 (−2.28 vs SFT) and raises the code-action share to 40.0%; step balancing keeps it at 26.1%, close to the SFT policy's 27.9%. RL without SFT reaches only 48.20.*

Headline Numbers

66.44%
OSWorld-Verified success
Hybrid baseline: 46.50%

- **63.87%** WinAgentArena
- **62.75%** 27B after SFT only
- **76.44%** Pass@3 RL; SFT 70.88

The paper's own controls

CONTROL (OSWORLD, 27B)	OS-HELIX-27B	COMPARED WITH
Matched Hybrid baseline	62.75 SFT / 66.44 RL	Qwen3.5-27B Hybrid 46.50
Nine OSWorld domains removed	62.31 (~17K samples)	62.75 (~31K samples)
Code access off, same policy	52.13	62.75 with code access
OSWorld-G / SS-V2 (offline)	32B: 57.21 / 93.47	Qwen3-VL-32B 61.35 / 95.05

Offline grounding is lower.

Takeaway

Hybrid SFT on verifiable synthesized tasks gives most of the gain; Step-balanced RL adds a step on one 27B model.

SFT: the main gain | RL alone: 48.20 | RL on 27B only | Evaluators imperfect

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Paper | Code